

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1118

Redbed Plains Wind Farm 138 kV
March 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1118. The Interconnection Customer proposes a 84.5 MW solar photovoltaic facility interconnecting at the Redbed Plains Wind Farm 138 kV substation in the Virginia area. The request was received on November 29, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 400 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,467,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SOUTH MEADOW 2 to SHORTPUMP 138.0 kV	LEAVITT to CHANCELLOR ROW SUBSTATION 345.0 kV Line	121.4	257	5.3
LEAVITT to S MEADOW 2 SUBSTATION 138.0 kV Cir. 2	LEAVIT to SOUTH MEADOW 2 138.0 kV line	118.3	341	15.4
PIPERS GAP to MESA 138.0 kV	LEAVITT SUBSTATION to CHANCELLOR ROW 345.0 kV L	117.7	190	5.3
STRATHMORE to HUMACAO 2601 138.0 kV L	CHANCELLOR ROW to LEAVITT 345.0 kV	117.7	189	6.9
SOUTH MEADOW 2 to LEAVITT 138.0 kV Line 1	SOUTH MEADOW 2 to LEAVITT 138.0 kV Line 2	116.7	345	15.4
SHORT PUMP to MESA 138.0 kV L	LEAVITT to CHANCELLOR ROW 345.0 kV	116.0	216	5.3
CHIPOLA to STRATHMORE 138.0 kV	CHANCELLOR ROW to LEAVIT 345.0 kV	115.6	214	6.9
DPS OCEAN SPRAY to WINDSOR HILLS 138.0 kV L	SHORT PUMP SUBSTATION to S MEADOW 2 138.0 kV	114.9	174	5.7
PIPERS GAP to DPS OCEAN SPRAY 138.0 kV L	LEAVITT to CHANCELLOR ROW 345.0 kV	111.3	179	5.3
SOUTHWICK to SOUTH MEADOW 2 138.0 kV	SHORT PUMP to S MEADOW 2 138.0 kV line	110.6	226	5.7
SOUTHWICK to WINDSOR HILLS 138.0 kV	SHORT PUMP to S MEADOW 2 138.0 kV	107.7	210	5.7
RIV RD to HUMACAO 2601 138.0 kV	LEAVITT to CHANCELLOR ROW 345.0 kV	103.5	188	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SHORT PUMP to S MEADOW 2 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,742,000	52% (\$1,953,000)
LEAVITT to SOUTH MEADOW 2 138.0 kV Line 2	Rebuild line with higher-capacity conductor	\$1,433,000	25% (\$363,000)
PIPERS GAP to MESA 138.0 kV	Rebuild line with higher-capacity conductor	\$3,210,000	52% (\$1,676,000)
STRATHMORE to HUMACAO 2601 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,746,000	10% (\$165,000)
LEAVIT to S MEADOW 2 138.0 kV Cir. 1	Rebuild line with higher-capacity conductor	\$959,000	25% (\$243,000)
SHORT PUMP to MESA 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,662,000	52% (\$1,911,000)
CHIPOLA to STRATHMORE 138.0 kV	Rebuild line with higher-capacity conductor	\$2,373,000	10% (\$225,000)
DPS OCEAN SPRAY to WINDSOR HILLS 138.0 kV	Rebuild line with higher-capacity conductor	\$6,171,000	52% (\$3,185,000)
PIPERS GAP to DPS OCEAN SPRAY 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,418,000	52% (\$1,784,000)
SOUTHWICK to SOUTH MEADOW 2 138.0 kV	Rebuild line with higher-capacity conductor	\$1,477,000	52% (\$762,000)
WINDSOR HILLS to SOUTHWICK 138.0 kV	Reconductor line	\$4,185,000	52% (\$2,160,000)
HUMACAO 2601 to RIV ROAD 138.0 kV	Reconductor line	\$421,000	10% (\$40,000)
	Total allocated to project		\$14,467,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.